

DE.AUTO

CHAPTER 1

HOW A CAR WORKS

THE WORLD OF CAR MODIFICATIONS MADE SIMPLE

Cars explained in plain, friendly English



POWERED BY DE.AUTO

DE.AUTO — Because knowledge is power, and cars are passion.

CAR MODIFICATIONS GUIDE MUST-KNOW TERMS

Car modifications, explained

Chapter One: How a Car Works

“EVERY JOURNEY STARTS UNDER THE HOOD.”

— DE.AUTO

Important notice

This guide is **entertainment only**. It is not educational or scientific content, and it is not a how-to manual. DE.AUTO is not an accredited scientific, engineering or technical authority. Everything here is simplified for fun and general interest, and it can differ from one car to another.

You must **never** try any modification, drive at high speed, or do any kind of stunt or dangerous driving on public roads. Safety and respecting traffic laws are everyone's own responsibility.

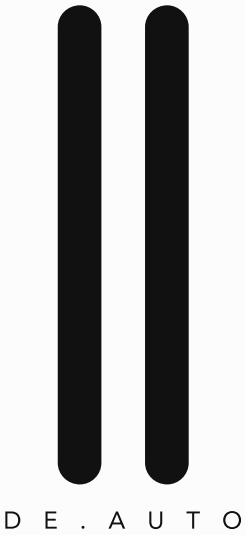
The information here is **general and approximate**, and much of it describes only the typical case. Laws, regulations and specifications differ by country and model, so always check the owner's manual and the official authorities.

Vehicle modifications are subject to local laws. Any unauthorized modification can lead to legal liability, fines or having your vehicle impounded.

DE.AUTO is not responsible for anyone who applies the information in this guide in real life.

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For any decision about your car, go to certified professionals and the manufacturer's manual, and only try something if it complies with official regulations and legal requirements. 🚀



DE.AUTO is more than a name. It's an identity that brings the passion for cars together with knowledge and progress.

At DE.AUTO we make fresh content for car lovers, mixing fun and know-how from the world of modifications and high performance, in a simple style that's easy to follow. Our goal is to help you understand your own car better and make smarter calls on modifications and maintenance.

We walk you through the key terms and parts of the modification world, so you can chat with the enthusiasts and follow every detail without feeling like an outsider.

Whether you're a beginner or a pro, you'll find terms, basics and fun facts here that help you get into cars in an organized, confident way.

DE.AUTO — Because knowledge is power, and cars are passion.

READY?

You drive your car every day... but do you know what happens under your foot the moment you hit the gas? 🤔

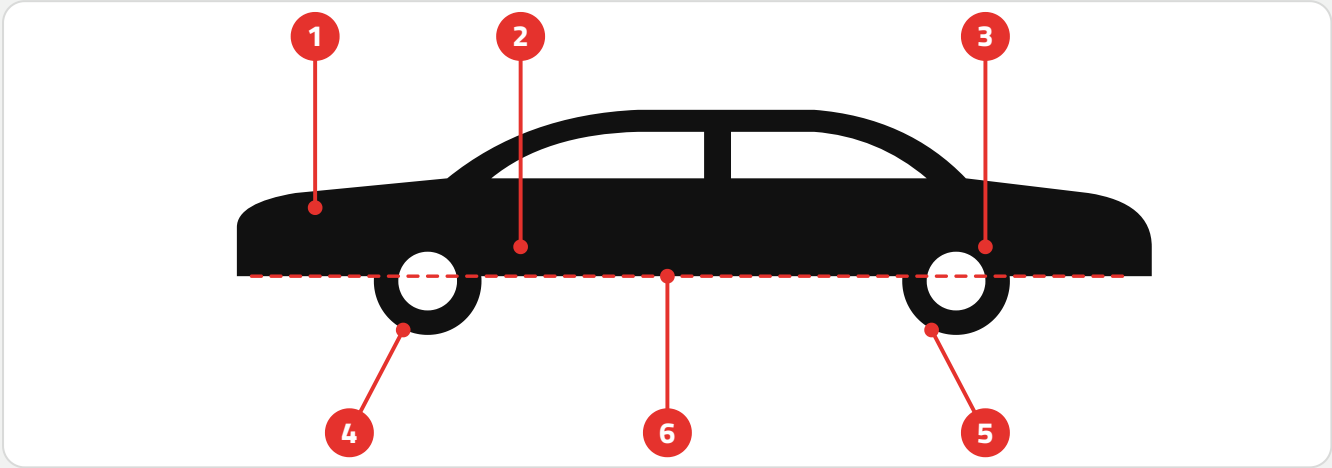
By the end of this chapter you'll know:

- ✓ How a car moves, and which parts work together
- ✓ How the engine makes power in four steps
- ✓ How that power gets from the engine to the ground (gearbox and differential)
- ✓ How to read your tire size (like 245/40R19)
- ✓ How brakes stop a ton of metal in seconds
- ✓ What runs your car's electrical system



How does a car move? 🚗

A car is a team of six systems, and each one hands off to the next. This is the map you'll follow through the whole series:



Simplified drawing. Part positions differ from car to car

1. Engine
Burns fuel and makes the power.

2. Gearbox
Sends the power from the engine to the wheels.

3. Differential
Splits the power between the wheels and lets them spin at different speeds in corners.

4. Tires
The only thing that touches the road.

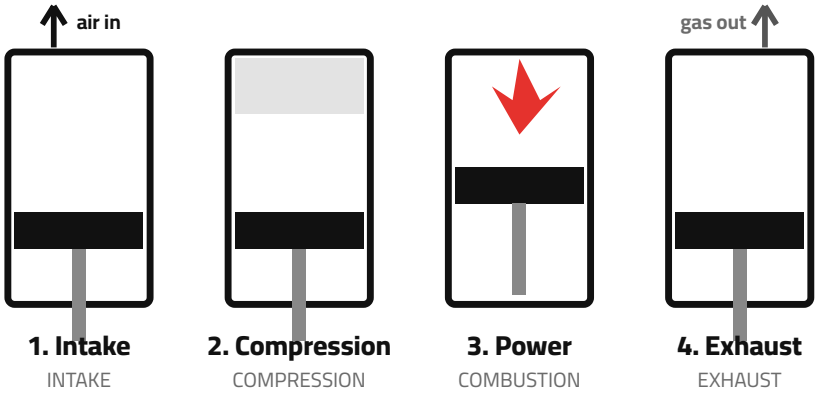
5. Brakes
Turn motion into heat through friction, so the car stops.

6. Electrical
Battery, alternator and computer. They start the engine and control it.

🔑 **The golden rule:** If you get these six, you get 90% of any conversation in the modification world. Every chapter in this series opens one of them up for you.

The engine: the beating heart

Almost every petrol engine does the same dance: **"suck, squeeze, bang, blow."** Repeat it thousands of times a minute and you get a car that moves. 🔁



The four strokes, left to right

1. Intake

The piston moves down and pulls in air. In most cars the fuel comes in with it.

2. Compression

The piston moves up and squeezes the mixture inside the cylinder.

3. Power

The spark plug fires, the mixture explodes and shoves the piston down. This is where the power is made!

4. Exhaust

The piston moves up again and pushes the burnt gases out of the cylinder.

15,000 RPM

revolutions per minute is what some racing engines reach, while normal road-car engines usually spin somewhere between 6,000 and 7,000.

💡 **Did you know?** Some racing engines use a Dry Sump system instead of the normal oil system, so oil pressure stays steady even in hard corners.

🤔 How does the engine time its valves to make even more power? We'll get into it in **Chapter 2: The Engine**!

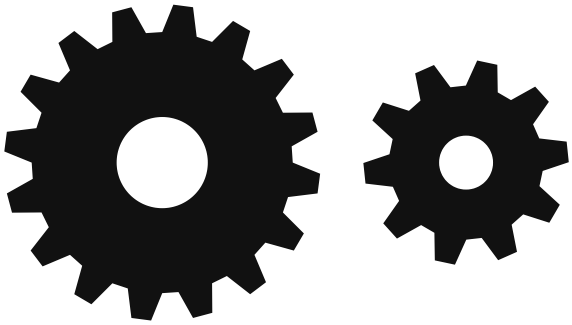
How does the power reach the wheels?

The engine spins, but what makes the wheels spin? That's where the **gearbox** and the **differential** come in.

The gearbox (Transmission)

Manual gearbox: you change gear yourself with the clutch.

Automatic gearbox: it changes gear on its own depending on your speed.



High gear = top speed **Low gear = pulling power**

🔢 **What's a «gear»?** It's a set of cogs that decides how much pulling power and how much speed reach the wheels. First gear gives big pulling power and low speed. Every time you go up a gear, speed goes up and pulling power goes down.

✕ MYTH	✓ FACT
An automatic is always slower than a manual.	That used to be true. Modern automatics (especially 8 gears and up, and dual-clutch DCT boxes) often accelerate as fast as a manual, or faster. We cover it in **Chapter 5.**

The differential (Differential)

In a corner, the outside wheel travels a longer distance than the inside one, so it has to spin faster. The differential is what lets the wheels turn at different speeds. That keeps the car from skidding and improves control.

💡 **Did you know?** Some four-wheel-drive cars have lockable differentials (Locking Differentials) that help off-road. We explain them in **Chapter 6: Drivetrain & Torque**.

Tires: your only point of contact

4 patches, each about the size of your palm. That's all that connects your car to the road! They're called the Contact Patch. All of your engine's power and all of your braking force go through them.

How do you read a tire size? 🔍

It's the number printed on the side of the tire, for example:

245/40R19

245	40	R	19
Tire width in millimetres	Sidewall height as a percentage of the width	Radial tire	Wheel diameter in inches

Air pressure: the right number is on the sticker in the driver's door. Check it when the tires are cold. Too little or too much weakens grip and hurts braking and stability.

Wear: every tire has a wear marker called TWI that shows you when it's time to replace it.

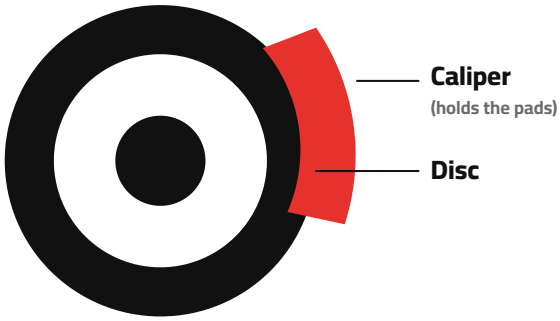
Age: a tire gets old and hard even if you never use it. The DOT code on the sidewall has the week and year it was made.

✕ MYTH	✓ FACT
A wider tire is always better.	A wider tire gives more grip on dry roads, but it can float more on water and it drinks more fuel. The best tire is the size that fits your car and the way you drive.

> 💡 **Did you know?** A bigger wheel with a thinner tire looks better and feels sharper, but it cuts ride comfort and is easier to damage on potholes. We go deeper in **Chapter 10: Tires & Wheels**.

How does a car stop? 🛑

Brakes don't "hold" the car. They **turn its motion into heat** through friction. The faster you go and the heavier your car, the more heat they have to handle.



Simplified drawing

Brake discs (Rotors)
Metal discs that spin with the wheel.

Brake pads
They squeeze onto the disc and slow the wheel.

Caliper
Holds the pads and presses them onto the disc when you push the pedal.

Brake fluid
Carries the pressure from the pedal to the caliper.

What happens when you push the pedal?

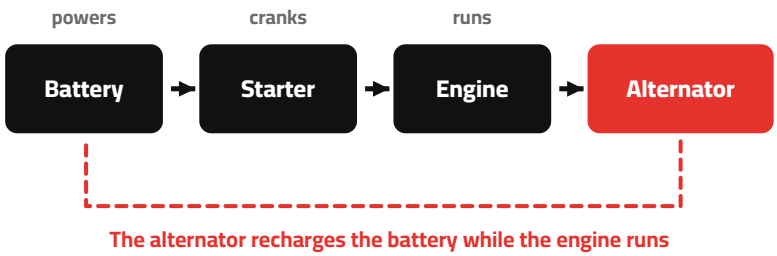
1. Your foot pushes the pedal, which pushes on the fluid inside the brake lines.
2. The fluid carries that pressure to every wheel (hydraulic pressure).
3. The caliper squeezes the pads onto the disc.
4. Friction slows the wheel, and the car stops.

> 💡 **Did you know?** Brake fluid absorbs moisture over time, and that cuts how well it handles high heat. That's why it gets replaced on the service schedule even if it still looks clean.

➡️ Safety systems like ABS and ESC are coming in **Chapter 9: Brakes & Safety**.

Car electrics: the brain and nerves

An engine doesn't run on fuel alone. Behind it is an electrical network that starts it and controls it:



Battery
Stores electricity and powers the starter and gadgets when the engine is off.

Alternator
Runs off the engine, charges the battery and feeds the electrics while you drive.

Starter motor
A motor that turns the engine over the first time, until it runs on its own.

Spark plugs & coils
Turn battery power into a strong spark that lights the mixture.

Computer (ECU)
Reads the sensors and controls the injectors, the spark and more.

Fuses
Protect the circuits and blow if there's an overload.

✕ MYTH
The battery runs the car the whole time.

✓ FACT
The battery only powers the starter. Once the engine is running, the alternator feeds the electrics and charges the battery.

💡 **Did you know?** Even electric cars have a small **12-volt** battery for lights and computers, separate from the high-voltage battery that moves the car.

🧠 **For mods:** The computer (ECU) decides how the engine behaves. In **Chapter 12: Car Tuning** we explain how its programming changes.

🎯 Test yourself!

Four quick questions. Answer without flipping back through the pages. The answers are on the last page.

🎯 Challenge 1 **Which part is the only one that actually touches the road?**

A) Brakes B) Tires C) Differential

🎯 Challenge 2 **In which stroke is the power made inside the engine?**

A) Intake B) Compression C) Combustion

🎯 Challenge 3 **In the size 245/40R19, what does the number 19 mean?**

A) Tire width B) Wheel diameter in inches C) Sidewall height

🎯 Challenge 4 **What recharges the battery while you drive?**

A) Starter B) Alternator C) Fuses

✅ **DE.AUTO tip:** If you got 4 out of 4, you're ready for Chapter 2. If not, flip back to that page. It won't take you a minute. 😉

Wrap-up 📌

Now you have the full picture: from the engine, to the gearbox and differential, to the tires, brakes and electrics. You know how your car moves! 🚀

🎯 Challenge answers:

1) Tires.

2) Combustion.

3) Wheel diameter in inches.

4) The alternator.

➡️ In **Chapter 2: The Engine**, we go deep inside: the difference between the block and the cylinder head, the types of engines, and the difference between horsepower and torque.